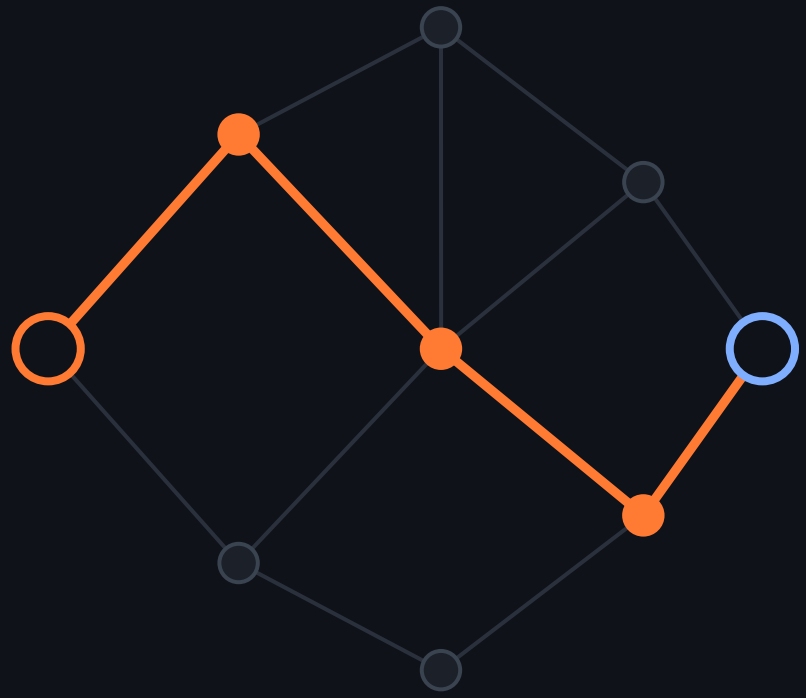
ZENTRAL HACK 2026 · AGENT HARNESS

Spec in.
Strategy out.



An agent harness that writes, verifies and evolves game strategies, for games it has never seen.

[Team name] · [names]

THE TASK · THE INSIGHT

An unseen game, a fixed budget, only wins count

The task

Read a game's written spec, ship a strategy that wins. Unattended, in a sandbox.

A **$0.03 · 3 min**

B **$0.50 · 10 min**

C **$10 · 30 min**

Scored on two unseen games: win +1, draw 0, loss −1.

What research says

85% vs **30%**

Win rate when the LLM writes a simulator plus search code, versus the LLM playing move by move.

Code World Models, Google DeepMind

Our insight

Make the LLM the programmer, not the player.

And never trust its code before it is verified.

Sources: Zentral Hack 2026 task description; Code World Models for General Game Playing, arXiv 2510.04542.

THE HARNESS

Many drafts, one verified simulator, one champion

`spec.md` in

`strategy.py` out

1

Write

3–4 strategy drafts with different personas, and a simulator, in parallel

2

Verify

The simulator must reproduce the spec's worked examples; drafts are probed and repaired

3

Self-play

Round robin on the simulator: both seats, many seeds, real time limits

4

Improve

The strongest model writes challengers against the champion's weak spots

5

Guard

Hard timer, fallback strategy, output check: a forfeit is impossible

↻ repeat while time and money last

Budget Every call reserves its worst-case cost before it is sent: it cannot overspend.

Deadline The clock counts from container start; the file on disk is valid at every second.

HOW WE BUILT IT

Built like it works: measure, decide, repeat

8

games: 2 dev, 6 our own, 2 of them held out

58

harness runs, every one measured

≈ $21

API cost in total

Claude Code as the engineering team: parallel agents wrote and reviewed every module, held to 216 automated tests.

EPOCH 1

Ablation · 24 runs

Pipeline ≫ one-shot. Improvement rounds: the biggest gain. The sparring draft hurt selection: removed.

EPOCH 2

Tournament conditions · 18 runs

Self-play now runs with the real time limits. Spare-time seeds and medium-effort improvement: kept.

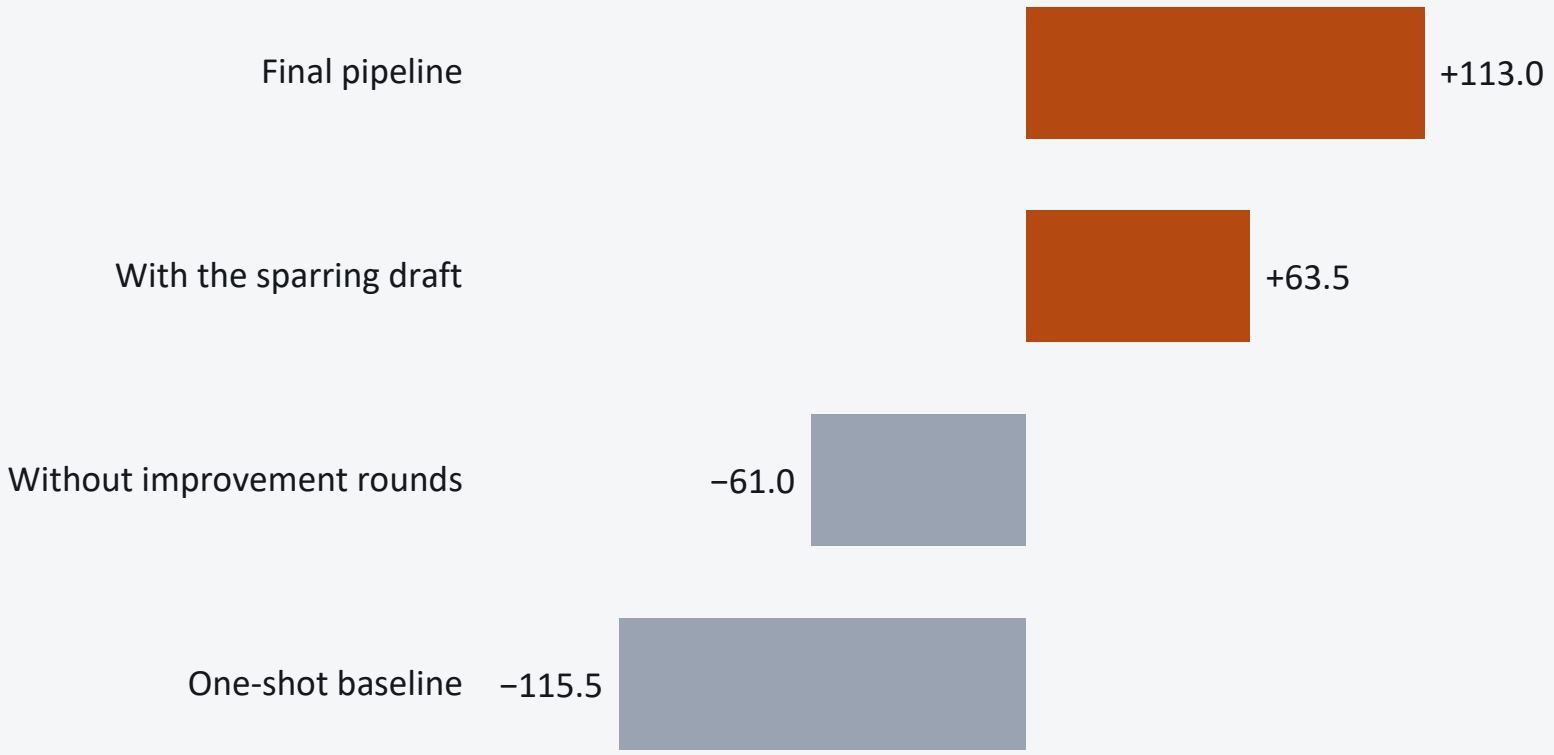
EPOCH 3

Failure mining · every log

Two agents mined all runs: 3 defects fixed, 6 ideas queued, each to pass a with/without test.

EPOCH 1 · RESULT

Every feature had to earn its place



48–0–0

pipeline vs one-shot, head-to-head on beta (gamma: 46–1–1)

2 of 3

games lost by each feature we tried to remove in epoch 2

Net points per strategy (win +1, draw 0, loss −1 per match), summed over 3 games: cross-tournament of the variants, both seats, 6 seeds.

THE EXAM

It transfers to games it was never tuned on

44–1–3

vs one-shot on a held-out game with hidden demand, random surges and typed units

32–0–16

on a held-out game with renamed fields and nested-dict actions

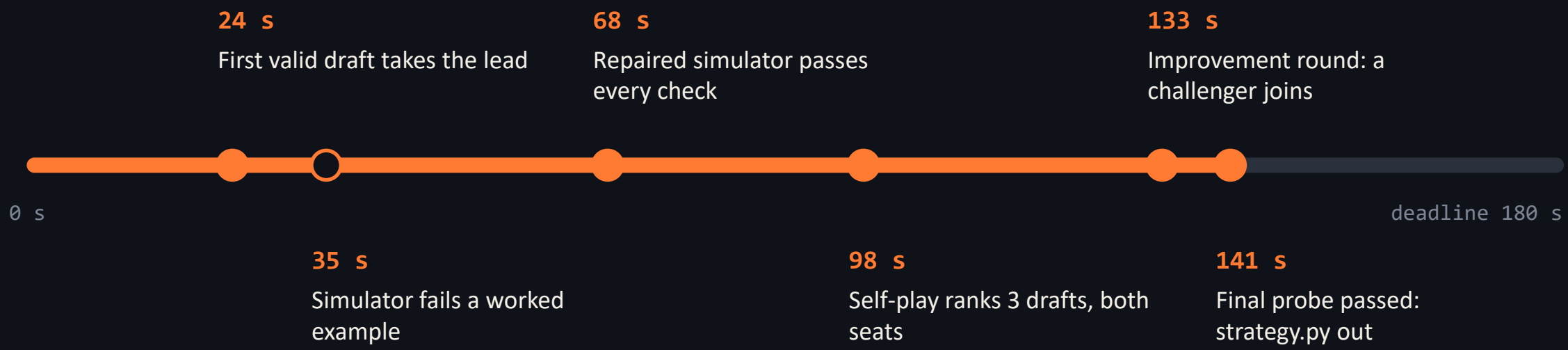
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delivered strategies pass the arena conformance check: zero forfeits

Task C check (sol + astra, 15 min): 100% of matches won against the baselines.

DEMO · ONE REAL RUN

141 seconds, 1.8 cents, one champion



Spec in. Strategy out. Never a forfeit.

Task A profile in the evaluation’s Docker container: $0.018 of $0.03.